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Incorporating Methane Isotopologues Alters Tropical and Subtropical Methane Emission Estimates

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Abstract

Annual increases in atmospheric methane have reached record highs over the past five years, yet the underlying processes and regional sources remain uncertain. Here we quantify the average 2019-2021 global methane budget averaging using Bayesian 4D-Var inversions that assimilate satellite methane retrievals and in-situ $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ measurements. The methane-isotopic inversion yields total emissions of 623 [585–643] Tg/y, slightly higher than the methane-only inversion. Incorporating isotopic constraints leads to a redistribution of emissions in tropical and subtropical regions. Compared with the methane-only inversion, the methane-isotopic inversion indicates that emission estimates increase by 26 [–3–27] Tg/y in East Asia (primarily China), 7 [–1–8] Tg/y in South Asia (primarily India), and 5 [–3–10] Tg/y in central Africa, while decreasing by 5 [–6–7] Tg/y in Amazon Basin and 12 [–14–20] Tg/y elsewhere. This points to a stronger anthropogenic contribution to the post-2019 methane budget, including higher fossil emissions in China and slightly less wetland emissions in the Amazon Basin. The methane-isotopic inversion also alters inferred emission seasonality, showing less seasonality in China compared with the methane-only inversion, weaker coal-mine phase-out signals, and a delayed summer peak in Southeast Asia, pointing to sources missing from current inventories.

Introduction

Methane (CH_4), as a major component of natural gas, is often referred to as a "bridge fuel" in the transition from fossil-based to renewable energy systems. However, as a potent greenhouse gas, methane is responsible for $\sim 30\%$ of the global temperature increase since the Industrial era¹. Reducing methane emissions can effectively influence climate on a decadal time-scale due to its strong global warming potential (83 times that of carbon dioxide over 20 years) and its relatively short lifetime (12 years)¹. Despite the urgency to mitigate climate risks associated with methane, current estimates of the contemporary global methane budget retain persistent uncertainties in characterizing methane sources and sinks accurately. For example, the uncertainty range in the global emission estimates (669 [512–849] Tg/y) from the 2010–2019 Global Methane Budget (GMB)² bottom-up statistics/models is 337 Tg/y, equivalent to $\sim 50\%$ of the ensemble mean, and most of the GMB model estimates lack the ability to represent variations in methane isotopologues (e.g., $\delta^{13}\text{C}\text{-CH}_4$)³.

The urgency of addressing these knowledge gaps is underscored by the recent acceleration in atmospheric methane concentration growth. The growth rate of atmospheric methane doubled in 2020–2022 (15–18 ppb/year; Figure 1 and Supplementary Fig. S1) compared to the 2010s (averaged 8 ppb/y), but returned to 8–9 ppb/y annual increase rate after 2023⁴. Previous studies hold contrasting views on the causes of this 2020–2022 surge. Proposed contributing factors include a decline in the methane sink driven by reductions in OH concentrations (associated with NO_x- and/or VOC-driven chemistry, possibly related to emission changes during the COVID lockdowns), climate–emission feedbacks such as biogenic emissions that are enhanced by monsoon rainfall, and tropical wetland emissions that are intensified by global warming^{5–10}. However, it is unclear whether the methane budgets building on the above perspectives are consistent with evidence from methane isotopologues. We thus are unable to predict whether this accelerated methane growth rate will continue or recur—in part due to insufficient understanding of the relative contributions from natural and anthropogenic sources, and their geographic emission distributions^{3, 5–8}.

Isotopic measurements are powerful tools for identifying and distinguishing the underlying mechanisms of methane production and emissions. Specifically, methane isotopologues provide key constraints for differentiating biological (microbial), thermogenic, and pyrogenic sources. Despite some overlaps, each of these source categories has relatively distinct isotopic signatures reflecting their formation processes (Supplementary Table S1). Biogenic methane, produced mainly through microbial fermentation and CO_2 reduction under anaerobic conditions, has lower values in $\delta^{13}\text{C}$ and δD due to strong kinetic isotope effects during methanogenesis preferring the lighter isotopologues in the substrate molecules.

Thermogenic methane, formed by the thermal cracking of organic matter at high temperatures and pressures, and pyrogenic methane, generated during high-temperature biomass combustion, are more enriched (heavier) in ^{13}C and D. The related processes involve much smaller isotopic fractionation than microbial methanogenesis, and the heavier isotopologues are thus not preferentially excluded during formation. As a result, emissions from wastewater or livestock have lower $\delta^{13}\text{C}\text{-CH}_4$ than emissions from biomass burning or fossil fuel production³, while the spatial pattern of $\delta\text{D}\text{-CH}_4$ source signature has a close relationship with hydrology processes^{11, 12}. Simultaneously, $^{13}\text{CH}_4$ and $^{12}\text{CH}_3\text{D}$ last longer on average in the atmosphere compared to $^{12}\text{CH}_4$, due to smaller reaction rate constants (oxidation by OH, Cl, and O^1D).

The methane increases during 2006–2016 are accompanied by a 0.2–0.3‰ decrease in $\delta^{13}\text{C}\text{-CH}_4$ ^{4, 13}. Studies^{14–17} consistently identify tropic/subtropic regions and economic-emerging countries as the main areas of emission growth, including China, US, India, central Africa, and the Middle East. However,

these studies attribute the emission increases differently across sectors, such as tropical wetlands, waste, agriculture, and fossil fuels. This 0.2–0.3‰ decrease in $\delta^{13}\text{C-CH}_4$ has been used as evidence of a growing contribution from microbial sources, such as agriculture, wastewater, and wetlands, over the past two decades^{3, 18}. However, other $\delta^{13}\text{C-CH}_4$ based studies^{19–21} suggest that fossil fuel still contributes to approximately half of emission increases through 2016, possibly driven by coal mining emissions rising sharply since 2007, and attribute the observed decrease in $\delta^{13}\text{C-CH}_4$ to decreases in biomass burning emissions²². In addition, model studies³ of $\delta^{13}\text{C-CH}_4$ rule out significant decreases in OH concentrations as the main reason for the post-2016 methane increases, in contrast to climate-chemistry model²³ and box model analyses^{24, 25} that identify OH variability as one of the major contributors. Recent studies^{21, 26} also highlight the substantial sensitivity of total budget estimates to uncertainties in isotopic fractionation effects, such as the role of methane loss to Cl oxidation. More recently, during 2016–2021, $\delta^{13}\text{C-CH}_4$ continued to decrease by an additional 0.1–0.2‰¹³, offering valuable insights into the sharp rise in methane concentrations observed after 2019.

Here, we reconcile the above disparate findings through joint inversions of satellite methane and in-situ isotopic observations. Measurements of methane isotopologues provide valuable information on emission sector attributions, but are sparse and have large spatial or temporal observational gaps. Satellite methane column data offer near-global coverage and frequent revisit times. Combining satellite and isotopic measurements thus offers a unique approach to improve total emissions estimates spatially and temporally^{27, 28}. Recent advances in inversion techniques have made the joint analysis of satellite data and isotopic data more feasible. For example, researchers have coupled analytical inversions with box models or chemical transport models, albeit at coarse regional scales, to investigate long-term methane emission trends, and applied variational inversions to better account for the spatial gradients in $\delta^{13}\text{C-CH}_4$ ^{19–21, 29, 30}. We combine 01/2018–06/2022 satellite methane total column observations from the TROPOspheric Monitoring Instrument (TROPOMI; version 02.04.00) with in-situ $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ observations from National Oceanic and Atmospheric Administration (NOAA)/ Institute of Arctic and Alpine Research (INSTAAR), to quantify the global methane budgets. This study addresses a key question: What information can isotopologues ($\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$) bring to improve the estimates of methane emission magnitude and distributions post-2019?

Results and discussion

Do more tracers improve emission estimates?

We integrate space-borne observations of methane with ground-based measurements of methane isotopologues ($\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$) within a suite of inverse analyses. We apply the GEOS-Chem adjoint model in a 4D-Var inverse system, to simultaneously optimize methane total emissions and its isotopic ($\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$) source signatures from total emissions on a grid level ($4^\circ \times 5^\circ$). To further constrain fossil fuel emissions, we also incorporate ethane (C_2H_6) observations and optimize ethane-to-methane emission ratios, leveraging the fact that ethane is co-emitted with methane primarily from fossil sources. The inversions are based on a set of partial differential equations to solve the joint probability distributions of methane total emissions and sinks, isotopic source signatures, and ethane-to-methane emission ratios (see Supplementary Table S1, Supplementary Fig. S2, and Methods). In addition to a full inversion that incorporates all tracers, we perform a series of sensitivity inversions to evaluate the contribution of each observational constraint. These include (see Table 1): A1 (methane only), B1 (methane + $\delta^{13}\text{C-CH}_4$), C1 (methane + $\delta\text{D-CH}_4$), D1 (methane + ethane), E1 (methane + $\delta^{13}\text{C-}$

CH₄ + δ D-CH₄), and F1 (methane + δ^{13} C-CH₄ + δ D-CH₄ + ethane), all using fixed OH fields derived from full-chemistry model simulations. We extend these configurations by optimizing OH: cases A2–F2 simultaneously optimizing multi-year averaged OH, and cases E3–F3 optimizing annually OH concentrations, on a grid basis. The optimized OH fields closely match prior estimates from full-chemistry simulations, which have been shown to agree well with aircraft measurements of atmospheric background OH and CO⁵. The inversion results are generally comparable across these three treatments of OH cases, and the primary differences in these solutions are driven by the choice of assimilated tracers.

Assimilating δ^{13} C-CH₄ and δ D-CH₄ can improve inversion results. The inverse system developed here can simultaneously improve simulations of multiple tracers, including methane, δ^{13} C-CH₄, δ D-CH₄, and ethane (Supplementary Fig. S3–S8). We further evaluate the inversion results by comparing with methane observations from a global collection of in-situ methane measurements (ObsPack), independent from the assimilated data. All inversions yield <15 ppb model biases, much lower than the -89 ppb prior bias. Including one additional tracer (B1–B2, δ^{13} C-CH₄; C1–C2, δ D-CH₄; D1–D2, ethane; see Table 1) results in similar performance as the methane-only inversion (A1–A2), yielding to 8–15 ppb model biases. When including both δ^{13} C-CH₄ and δ D-CH₄ (E1–E3), the inversion yields the best performance (0–2 ppb bias, see Supplementary Fig. S3). However, inversions F1–F3, which further include ethane, show reduced performance (up to -12 ppb mean bias) compared to E1–E3.

Ethane and methane are co-emitted from oil and gas production, and ethane is commonly used as a secondary tracer to constrain methane emissions from this sector. However, as highlighted by a previous study³¹, uncertainties in the ethane:methane emission ratio and its spatial and interannual variability limit the effectiveness of using ethane to infer methane emissions. Moreover, the coarse model resolution (~400 km) in this study is not well suited for capturing localized emission processes, such as plumes from individual oil and gas facilities. This spatial mismatch may reduce the added value of assimilating ethane in a global framework.

Thus, considering the evaluation against the independent dataset, ObsPack, we refer to the ensemble mean of inversions E1–E3 (assimilate methane, δ^{13} C-CH₄, and δ D-CH₄, with fixed OH or optimized OH; hereafter referred to as the methane-isotopic joint inversion) as the base case, with uncertainties derived from the full range of all inversions (Table 1). These base inversions E1–E3 reduce root-mean-square-error by 70%–71% for methane (TROPOMI), 57%–58% for δ^{13} C-CH₄, and 28% for δ D-CH₄, relative to the assimilated observations. E1–E3 also reduce the model simulation mean bias by 88%–91% compared to the assimilated methane data and more than 98% compared to independent evaluation data (ObsPack). Considering that the abundance of δ^{13} C-CH₄ and δ D-CH₄ is low and the effect of isotopic fractionation is modest but cumulative, isotopic ratios equilibrate more slowly than total methane. Coupled with the sparse coverage of isotopic measurements, interannual variability in sectoral emissions is likely less well constrained than multi-year averages. Thus, we focus on the average methane budget across 2019–2021. Given the sparsity of δ D-CH₄ measurements, we construct a multi-year average (post-2004) diurnal climatology by averaging the available measurements, and analyze the multi-year averaged isotopic signatures. We analyze the δ^{13} C-CH₄ source signature for 2019, given the more complete observational data available that year.

The selected methane-isotopic joint inversions (E1–E3, assimilate methane, δ^{13} C-CH₄, and δ D-CH₄) improve the global methane budget estimates in three ways, they:

- 1) Address part of the biases in the global total emission and sink estimates. Isotopologues ($^{13}\text{CH}_4$, CH_3D) differ from $^{12}\text{CH}_4$ as a result of their characteristic emission source signatures and kinetic isotope effects. The joint inversions leverage these constraints to solve the joint probability distributions of sources and sinks. This results in a small increase in global emission (3%, +21 Tg/y; discussed later) and in global sink (1.4%, 7.8 Tg/y) estimates, leading to a net increase in the inferred global source–sink discrepancy. The methane-only inversions reproduce TROPOMI column methane well but exhibit a -9 to -10 ppb bias compared with ObsPack, likely due to retrieval biases or representativeness differences. By incorporating isotopic in-situ measurements, the inversions gain an additional, independent dimension of information that helps reconcile the inconsistencies between satellite and surface observations. After assimilating isotopic data, the posterior simulations show a much smaller bias (within 2 ppb) relative to ObsPack.
- 2) Improve estimates of emission spatial distributions and temporal variation. As discussed earlier, isotopic measurements are sparse in space and time, but this limit is partially offset by the near global coverage of satellite observations. The 4D-Var inversion integrates these diverse data while accounting for realistic atmospheric transport, enabling a more consistent reconstruction of source distributions, thus improving both the spatial and temporal variability of inferred methane emission estimates. As we show in later sections, these improvements play a central role in resolving emission redistributions across the tropical and subtropical regions, where observational gaps and prior inventory biases are pronounced.
- 3) Improve source attribution to sectors. The inversions optimize emission-weighted isotopic signatures of total emissions, which can blend real sectoral shifts with prior source signature biases and model transport errors. To disentangle these effects, we further refine the inverse solution into sector level through a new approach. This refined approach identifies the optimal solution by iteratively adjusting inventory-based sector fractions to maximize consistency with the optimized source signatures, while preserving grid-level total emissions from the 4D-Var solution (see Methods). This approach isolates what sectoral emission fractions would need to change to be consistent with the isotopic signals and the optimized total emissions. The refined emission corrections have lower spatial correlations across sectors, indicating improved separation among fossil, livestock, waste, and other anthropogenic emissions (Supplementary Fig. S9).

The optimized 2019–2021 global methane budget that is consistent with satellite and isotopic observations

The optimized global methane budget for 2019–2021 from the methane-isotopic joint inversions includes total emissions of 623 [585–643] Tg/y and sinks of 572 [533–590] Tg/y (accounting for both atmospheric oxidation and soil uptake). The inferred emissions exceed prior bottom-up inventory estimates (517 Tg/y) by 106 [68–126] Tg/y, and exceed the methane-only inversion estimates (602 Tg/y) by 21 Tg/y. They also exceed non-isotopic inversions from a previous study of 2020–2021 (587–592 Tg/y)³², but lies between the bottom-up inventories (669 [512–849] Tg/y) and top-down (575 [553–586] Tg/y) model ensembles from the Global Carbon Project’s (GCP) 2010–2019 emission estimates.

Attributing emissions to individual sectors reveals contributions of 110 [87–112] Tg/y from fossil fuels, 137 [131–140] Tg/y from livestock, 92 [88–94] Tg/y from wastewater and landfills, 34 [27–34] Tg/y from rice cultivation, and 30 [28–31] Tg/y from other anthropogenic sources (such as industrial combustion). Natural emissions account for an additional 177 [171–206] Tg/y from wetlands (including freshwater) and 43 [37–46] Tg/y from other natural sources, as summarized in Supplementary Fig. S10.

Compared with the methane-only inversion, the methane-isotopic joint estimates show the largest upward emission revision in the sector of fossil fuel (+21 Tg/y), followed by rice cultivation (+6 Tg/y), natural sources other than wetlands (+5 Tg/y), livestock (+4 Tg/y), waste (+2 Tg/y), and other anthropogenic sources (+2 Tg/y). In contrast, wetland emissions are revised downward by -19 Tg/y. For comparison, studies of pre-industrial $\delta^{14}\text{C-CH}_4$ suggest that global anthropogenic fossil emissions are greatly underestimated in inventories, resulting at >150 Tg/y fossil emissions and 2 Tg/y seep emissions during 2003–2012³³. Our 2019–2021 estimate of anthropogenic fossil sources (110 [87–112] Tg/y) is lower than this 2003–2012 estimate. However, both our results and this pre-industrial $\delta^{14}\text{C-CH}_4$ study³³ point to an underestimated role of anthropogenic fossil compared to the prior inventory (80–83 Tg/y).

Consistent with sectoral emission attribution, the optimized global-averaged emission-weighted source signatures of $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ are heavier than their prior estimates. The global-mean emission-weighted source signature of $\delta^{13}\text{C-CH}_4$ is -53.74‰ in 2019, heavier than the prior estimate of -55.96‰, though still lighter than -52.5‰, the 2012–2014 top-down estimates from a previous study²⁰. Similarly, the global-mean emission-weighted source signature of $\delta\text{D-CH}_4$ during 2019–2021 is -261.27‰, also heavier than the prior estimates of -288.60‰. The methane-isotopic joint inversions require a heavier global source signature to match atmospheric $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ measurements, resulting in a higher contribution of fossil emissions to the total methane budget than prior inventories and methane-only inversions suggest. The optimized $\delta^{13}\text{C-CH}_4$ source signatures (Supplementary Fig. S11) exhibit globally heavier values relative to the prior, yet reveal lighter regional features, particularly in northern China and Russia, suggesting enhanced biogenic emissions or revised source mixing in those regions. In contrast, the optimized $\delta\text{D-CH}_4$ signatures (Supplementary Fig. S12) highlight several previously underrepresented regions with anomalously heavy source signatures: the US Upper Midwest (near the Great Lakes), eastern China (near Shanghai and Beijing), the maritime region spanning Malaysia, Indonesia, and Papua New Guinea, and northern India. These spatial anomalies indicate potential unidentified or underestimated isotopically enriched methane sources, warranting further investigation into regional fossil fuel production, biomass combustion, or geologic emissions not captured in current inventories.

Since 2007, the global-mean atmospheric $\delta^{13}\text{C-CH}_4$ has shown a decreasing trend (heavier). This trend has been widely interpreted as evidence that microbial sources have played a dominant role in the post-2007 methane increase, in contrast to the pre-2006 period, or potential changes in isotopic fractionation by sinks³⁴. A recent work³⁵ also attributes the early-2020 methane surge to enhanced microbial emissions and a reduction in OH. Our optimized source signatures (and the inferred larger role of fossil sources) are not in conflict with this long-term atmospheric trend. Rather than reflecting changes relative to the previous decade, our results quantify deviations from prior inventory-based estimates for the same period (i.e., 2019–2021). This does not preclude a long-term increase in light-isotope emissions; instead, it points to a gap that the emission sector fractions in current inventories cannot reproduce the observed isotopic composition in atmospheric measurements.

Constraints from $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ redistribute methane emissions in tropical and subtropical areas: reduced wetland emissions in Amazon and enhanced fossil emissions in China

Even though assimilating $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ only slightly increases the global methane emission estimate by 3% compared to the methane-only inversion, the integrated approach suggests important emission spatial redistributions, particularly in tropical and subtropical regions. These regions have been

identified as major contributors to the recent acceleration in methane concentration increases². Compared with the methane-only inversion, incorporating isotopic constraints leads to reduced wetland emissions in the Amazon Basin and northern Latin America and enhanced anthropogenic emissions in Asia and Central Africa.

The regions where emissions have the largest upward revisions after we assimilate $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ are East Asia (primarily China, regions defined in Supplementary Fig. S13; +26 Tg/y, see Figure 2 and Table 2), followed by South Asia (primarily India; +7 Tg/y), and Central Africa (notably Nigeria and the Democratic Republic of the Congo; +5 Tg/y), compared with the methane-only inversion. The inversions suggest more enriched (heavier) isotopic source signatures for the total emissions than prior estimates in these regions: the increases in $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ source signatures for total emissions are +2‰ and +45‰ in East Asia, +6‰ and +34‰ in South Asia, and +6‰ and +39‰ in Central Africa, respectively. However, these emission enhancements are attributed to different source sectors. In East Asia, fossil fuel contributes an estimated 62% (+16 Tg/y) of the increases in emission magnitude, particularly in the North China Plain, up to +90 kg/day/km², and the Southwestern China (Yunnan–Guizhou–Sichuan). However, some northern China regions exhibit lighter $\delta^{13}\text{C-CH}_4$ source signatures, reflecting potential increases in both biogenic and fossil fuel emissions, with revised source sectoral mixing. In South Asia, agriculture is the major contributor, explaining 74% (+5 Tg/y) of the emission revisions, mainly in the northern India. While in Central Africa, the emission revisions are 42% from wetlands, 38% from fossil, and 21% from other sources.

Conversely, emissions over Brazil (-5 Tg/y, Figure 2 and Table 2), northern Latin America (-4 Tg/y), Southeast Asia (-2 Tg/y), and the rest of the world (-8 Tg/y), decrease when isotopic constraints are added. Wetlands account for most of the downward emission revisions, explaining nearly all of the -5 Tg/y emission revisions in Brazil and 68% (-2.5 Tg/y) in northern Latin America. The -5 Tg/y emission revision in Brazil is equivalent as 3% of the total wetland emissions over the Amazon Basin. In Southeast Asia, the emission revisions are a combined effect of natural emissions, -4.3 Tg/y downward revision from wetlands, offset by 3.4 Tg/y upward revision from other natural sources. The spatial patterns of these emission revisions are also mixed, showing upward revisions over southern Sumatra and Indonesia-Kalimantan, and downward revisions in northern Sumatra and Papua New Guinea. This suggests potential changes in wetland and freshwater distributions or substantial spatial biases in current wetland inventories.

In tropical and subtropical regions, satellite-based methane observations (e.g., from TROPOMI and Greenhouse Gases Observing Satellite (GOSAT)) are often limited by persistent cloud cover and high humidity, leading to data gaps (Figure 1). As a result, satellite-informed inversions tend to scale up emissions across large areas to reconcile with the background methane from limited observations, making the inferred emissions highly sensitive to the magnitude of OH, which itself carries substantial spatial and temporal uncertainty. Assimilating isotopic measurements ($\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$) adds a valuable layer of constraint by distinguishing between biogenic, thermogenic, and pyrogenic sources. This additional information helps to refine source attribution and thus improves the spatial accuracy of emission estimates. In future studies, enhanced satellite coverage in the tropics and improved OH estimates should further reduce remaining uncertainties.

Combining all the emission corrections as discussed above after implementing isotopic constraints, we present the optimized methane budgets for tropical and subtropical regions in Figure 3 and Table 2, and summarize these budgets as follows.

- 1) In East Asia, the optimized emissions total 91 Tg/y, with the largest contribution from fossil sources (39 Tg/y). It is the region where the inversions led to the greatest estimated emission increase, with prior inventories underestimating emissions by 26 Tg/y, and 16 Tg/y of this underestimation is from fossil (Figure 4).
- 2) In South Asia, the optimized total emission is 56 Tg/y, more than half of which is from agriculture (33 Tg/y). Agriculture emissions are also underestimated in the prior, by an estimated 8 Tg/y, mainly in the northern India, Nepal, and Bangladesh (Figure 4). This is the region where agricultural emissions are most intensively underestimated globally.
- 3) In Southeast Asia, the optimized emissions total 41 Tg/y, comprising wetlands (13 Tg/y), agriculture (10 Tg/y), and other sources (17 Tg/y).
- 4) Central Africa has an optimized total emission of 61 Tg/y. Wetland emissions in Congo rainforest are underestimated by around 9 Tg/y in the prior. Nigeria is another hotspot where emissions have been underestimated. While an earlier study³⁶ attributes the rising methane levels in Nigeria primarily to rice cultivation, our results point instead to fossil sources as the dominant cause.
- 5) For Brazil and Northern Latin America, the optimized emissions are 61 Tg/y and 26 Tg/y, respectively, dominated by wetland and agricultural sources. Although assimilating isotopic data reduces posterior emissions relative to the methane-only inversions, prior inventories still underestimate emissions by 9 Tg/y in Brazil, mostly occurring from November to May, corresponding to the rainy season. This suggests that improving seasonal wetland extent estimates may be able to reduce the bottom-up emission biases.
- 6) The inversions also identify a hotspot of underestimated fossil emissions in the Iraq-Kuwait-Iran region, accounting to +2 Tg/y in total, compared with prior inventories (Figure 4). This aligns with earlier studies^{5, 17} reporting increased fossil fuel activities in this region over the past decade.

Methane isotopologues ($\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$) indicate altered seasonal cycle of global and regional methane emissions

The seasonal cycle plays an important role in quantifying global and regional methane budgets. Emissions from wetlands, rice cultivation, manure, and wastewater respond to seasonal changes in temperature, precipitation, and inundation. Biomass burning adds further seasonal fluctuations. Atmospheric OH, the primary oxidant of methane, is photochemically produced and exhibits a pronounced seasonal pattern due to the availability of UV radiation. Recent studies^{37, 38} have relied on the seasonal phase and amplitude of methane concentrations to infer interannual variabilities of methane emissions, sinks, and transport processes. Inaccurately representing seasonal emission cycles can bias annual methane estimates and distort our understanding of how methane responds to environmental and meteorological changes.

Assimilating isotopic measurements in addition to methane suggests greater emissions from November to May that are not captured by the methane-only inversions, particularly in northern China. As shown in Figure 5a, global atmospheric methane concentrations rise sharply from June to October and more gradually from November to May. When assimilating only methane mole fraction data, the inversion attributes a 55 Tg underestimate in prior global emissions for June–October and a 29 Tg underestimate

for November–May. However, further assimilating $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ data (Figure 5b) results in a more balanced distribution of the emission underestimates: 49 Tg from June to October and 58 Tg from November to May. Compared to methane-only inversions, this shift (as shown in Supplementary Fig. S14) is characterized by -3 Tg reduced emissions in the Amazon Basin during the dry season (June–October, 89% reduced emissions are from wetlands), and enhanced emissions during November to May in East Asia (18 Tg; 55% from fossil fuel), South Asia (+8 Tg; 57% from livestock), and central Africa (+6 Tg; 66% from wetlands). The largest emission enhancement is over northern China. In comparison, a previous study³⁹ identified decreasing coal-mining emissions with increased rice cultivation emissions in China from satellite observations, and attributed this shift to coal-mine phase-outs under recent energy policies. However, rice cultivation exhibits a pronounced seasonal cycle, whereas coal mining is relatively seasonally flat. Our methane-isotope joint inversion indicates that incorporating isotopic constraints yields a much flatter seasonal emission profile over China than the methane-only inversions, with a heavier-than-prior source signatures for both $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$, reflecting an underestimate of coal mining (or other fossil) emissions. This suggests a weaker coal-mine phase-outs signal in the methane-isotopic joint assimilation than would be inferred from the non-isotopic inversion. Nevertheless, consistent with previous study³⁹, our inversions also identify an underestimated emission peak from October to December in southeast China and Bangladesh. This emission underestimate in China is attributed to rice. However, at the model resolution, rice and wetland emissions in Bangladesh cannot be reliably separated, and the resulting signal reflects combined biogenic emissions. Unlike the July–October rainfall-fed biogenic emissions identified in northern India by previous work⁵, the emission peak in southeast China and Bangladesh occurs during dry season and is influenced by agricultural management practices⁴⁰. This finding highlights the need for more detailed representations of anthropogenic carbon activities in Asian fast-developing countries to improve the accuracy of emission estimates.

Assimilating isotopic measurements in addition to methane also reveals a notable temporal shift in the seasonal emission peak. When only assimilating methane, global emissions peak in August at 1917 Gg/d, 32% higher than the prior estimates (1456 Gg/d). The methane-isotopic joint inversion indicates a lower August emission (1738 [1698–2153] Gg/d) and a delayed seasonal peak in September, with emissions rising to 1938 [1793–2093] Gg/d (for comparison, 1488 Gg/d in prior and 1887 Gg/d in methane-only inversion). Previous studies coupling an ensemble of wetland inventories with atmospheric observations also pointed to an autumn emission peak, but occurs in boreal regions⁴¹. In our solution, the delayed peak reflects longer-lasting emissions from tropical wetlands (e.g., Central Africa), earlier rice harvesting cycles in South and Southeast Asia, and underestimated September emissions in Malaysia and Indonesia. Compared to prior inventories, the emission underestimate in Southeast Asia is 146 Gg/d in September, accounting for roughly one-third of the global emission underestimation. Based on the prior emission fraction, these Southeast Asia emission underestimates arise mainly from natural emissions other than wetlands. As discussed previously, Southeast Asia shows a mixed pattern of spatial emission revisions, with decreases in wetland emission estimates offsetting by increases in other natural emissions. This pattern potentially indicates biases in the wetland/freshwater distributions in current inventories. However, the sparse observational coverage of methane, $\delta^{13}\text{C-CH}_4$, and $\delta\text{D-CH}_4$ over island and oceanic regions limits our ability to resolve sector-specific contributions, highlighting the need for additional data in the future.

The optimized $\delta^{13}\text{C-CH}_4$ emission-weighted source signatures are more correlated to seasonal temperature and hydrology variations than their prior estimates (Supplementary Fig. S15). The $\delta^{13}\text{C-}$

CH₄ emission-weighted source signatures correlated to soil temperature (0–7 cm) with a Pearson correlation coefficient $R = 0.88$, compared to $R = 0.72$ in the prior, and correlated to runoff with $R = 0.83$, compared to $R = 0.69$ in the prior. Biogenic methane emissions are strongly influenced by temperature and moisture, with higher emissions typically observed during warmer and wetter periods due to enhanced microbial activity. This sensitivity is particularly evident in ecosystems such as wetlands, wastewater treatment systems, and agricultural environments, including livestock operations and rice paddies. In contrast, pyrogenic sources, such as wildfires, are not directly temperature-sensitive in terms of methane production; rather, their occurrence and intensity are influenced by fire occurrence, which is influenced by heatwaves and droughts. Thermogenic sources, such as fossil fuel extraction, are primarily governed by anthropogenic activities and are less sensitive to short-term environmental variability. Thus, the stronger correlations observed between $\delta^{13}\text{C}$ -CH₄ source signature and temperature/runoff may reflect three potential scenarios:

- 1) Prior inventories may underestimate the sensitivity of biogenic emissions to environmental drivers. Current process-based models represent this sensitivity using process-specific parameters, such as temperature-dependence factor Q_{10} , but estimates of these parameters remain uncertain⁴². This is consistent with previous studies, which also demonstrated stronger sensitivities of biogenic methane emissions to rainfall and temperature changes than current process-based models^{2, 8, 10, 42}.
- 2) The results may indicate omissions or misclassifications of microbially derived methane. For example, in geological formations—such as coal seams or organic-rich shales—microbial methanogenesis can generate biogenic methane⁴³. However, these sources are typically classified as fossil in inventories and are often assumed to have negligible seasonal variability.
- 3) Atmospheric oxidation of methane by OH and Cl, both of which are temperature-dependent processes, can alter the isotopic composition of atmospheric methane. Uncertainties in the parameters of kinetic isotope effects (KIEs) may lead to seasonal biases in the inferred source signatures. In particular, although Cl plays a relatively minor role in the total methane sink, it has a disproportionate influence on $\delta^{13}\text{C}$ -CH₄, especially through seasonal oxidation in the marine boundary layer, where Cl forms from sea salt aerosols^{21, 26}. Improved estimates of Cl abundance and its KIE are essential for refining our understanding for the seasonal cycle of $\delta^{13}\text{C}$ -CH₄ and δD -CH₄. However, the major emission adjustments we identify occur over continental regions, where OH is the dominant oxidant. The sensitivity of our model to KIE of OH has limited seasonal dependencies, as explained in the Methods (also see Supplementary Fig. S16–S17). We thus consider this scenario less likely than the other two.

In summary, assimilating isotopic measurements in addition to methane reveals linked features in the seasonal cycle that are consistent with the spatial emission redistribution in tropical and subtropical regions. First, the methane-isotopic joint inversion shows enhanced emissions from November to May compared to the methane-only inversion, particularly over northern China. This aligns with the decreased wetland emission estimates in the Amazon Basin and the increased anthropogenic emission estimates across East Asia, South Asia, and Central Africa. Second, the methane-isotopic joint inversion identifies a shift in the timing of the seasonal emission peak, mainly due to emissions in Southeast Asia. This is consistent with the mixed spatial changes of source contributions in this region, including reduced wetland emissions offset by increases in other sources. Third, the $\delta^{13}\text{C}$ -CH₄ source signatures of total emissions exhibit stronger temperature and hydrological dependencies than the inventory-based prior estimates, demonstrating the potential of isotopic seasonal cycle to diagnose methane sources and

sinks in future studies. Taken together, these findings demonstrate that isotopic constraints not only refine the emission sector attribution, but also clarify emission seasonal pattern and regional drivers.

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Methods

Satellite and in-situ observations for methane, methane isotopologues, and ethane

We use satellite remote sensing methane total column measurements from TROPOMI version 02.04.00, which is based on S5P-RemoTec full-physics algorithm, with additionally albedo bias correction⁴⁴ and improved surface reflectance spectral dependencies⁴⁵. Launched in October 2017, TROPOMI embarked on its mission aboard the Copernicus Sentinel-5 Precursor satellite, orbiting Earth in a low-Earth polar trajectory. TROPOMI conducts near-global observations daily at 13:30 local time. It captures methane column measurements ($5.5/7 \times 7 \text{ km}^2$ nadir resolution, with a 2600 km swath width) utilizing a blend of near-infrared (NIR, $0.8 \text{ }\mu\text{m}$) and shortwave infrared (SWIR, $2.3 \text{ }\mu\text{m}$) spectral data. We exclude observations of latitudes above 60° or Quality Assurance (QA) value less than 0.5, for high-quality data that are not biased by high solar angles, high viewing zenith angles, or atmospheric/meteorological conditions. In total, 514 million measurements meet the quality standards across 01/2018–05/2022. We average these measurements to match model resolution ($4^\circ \times 5^\circ$), and a total of 930,340 data points remain viable for assimilation (Supplementary Fig. S18).

The $\delta^{13}\text{C}\text{-CH}_4$ dataset combines measurements from the NOAA collection^{21, 46} with additional data from the platform of World Data Centre for Greenhouse Gases (WDCGG⁴⁷). The NOAA collection includes data from multiple laboratories, with the largest contribution from the partnership between the NOAA Global Monitoring Laboratory and the Institute of Arctic and Alpine Research (INSTAAR) of University of Colorado at Boulder. To reduce biases arising from the use of different primary reference materials across laboratories, we apply inter-institutional corrections to the WDCGG data⁴⁸. All measurements are on the INSTAAR realization of the Vienna Pee Dee Belemnite (VPDB) scale. We filter the data using quality flags to remove records with known sampling problems, poor-quality flask pairs, and outliers. A total of 2589 measurements remain viable for the assimilation across 01/2018–12/2021, with denser coverage during 01/2018–06/2020 (Supplementary Fig. S18). The remaining data are aggregated into monthly means for use in the inversion.

The $\delta\text{D}\text{-CH}_4$ dataset combines measurements from INSTAAR⁴⁹, Institute for Marine and Atmospheric Research Utrecht (IMAU^{50, 51}), and the data collection platform WDCGG⁴⁷. We apply inter-institutional primary reference material bias correction⁴⁸ and filter the data using quality flag to exclude records with known sampling problems. The offset between the INSTAAR and MPI scales is 11.5‰. Given the sparsity of $\delta\text{D}\text{-CH}_4$ measurements, we construct a multi-year average (2004–2017) diurnal climatology by averaging the available measurements (Supplementary Fig. S18). A total of 9869 measurements remains viable for assimilation.

Measurements of ethane are from NOAA ground-level observation networks⁴⁹ and ACT-America⁵². We filter the data by quality flag following the similar process as methane isotopologues. We select ethane measurements at locations where fossil fuel contributes to $\geq 30\%$ total methane emissions. In total, 2589 data points remain viable for assimilation.

We use independent measurements of methane from ObsPack v5.1⁵³ to evaluate the optimized methane budget. This product compiles 429 atmospheric methane datasets, including measurements from 54 laboratories across 22 nations. Thus, it is a comprehensive and valid evaluation database for our global analysis.

Forward model simulation for methane and its isotopologues

We use the 4D-Var inverse framework in GEOS-Chem adjoint model v35^{5, 27, 54, 55}, with $4^{\circ} \times 5^{\circ}$ spatial grids and 47 vertical layers. The forward model integrates GEOS-FP meteorological data provided by NASA's Global Modeling and Assimilation Office⁵⁶, operating on 5-minute intervals for transport and 10-minute intervals for emissions. For comparisons between simulations and observations, we sample model outputs as aligned with observation times (or satellite overpass time) and locations.

We combine state-of-the-art bottom-up inventories for the prior estimates of methane emissions. Global anthropogenic emissions are from the Emissions Database for Global Atmospheric Research (EDGAR) v6⁵⁷, superseded by the Updated Global Fuel Exploitation Inventory (GFEI v2⁵⁸) for fossil fuel emissions. We also supersede it by regional inventories, including Scarpelli inventories for Canada⁵⁹ and Mexico⁶⁰, and Gridded EPA inventory for US⁶¹. Global natural emission estimates include the ensemble mean of WetCHARTs⁶² for wetlands, Etiope⁶³ for geological seeps, and FUNG⁶⁴ for termite. Biomass burning emissions are from the Quick Fire Emissions Dataset (QFED⁶⁵). For the year 2020, we scale down emissions on source sectoral level to reflect the effect of COVID lockdowns, based on scaling factors in an earlier study⁶⁶. The prior estimate of global methane emission is 516–517 Tg/y during 2019–2021, including 80–83 Tg/y fossil fuel, 121 Tg/y livestock, 80–81 Tg/y landfill/waste, 27 Tg/y rice, 26–27 Tg/y other anthropogenic emissions, 147–148 Tg/y wetlands, and 33 Tg/y other natural emissions (include biomass burning).

We synthesize state-of-the-art estimates of methane isotopic source signatures, as shown in Supplementary Fig. S19–S20 and Supplementary Table S1. Prior estimates of $\delta^{13}\text{C-CH}_4$ source signatures incorporate anthropogenic and termite estimates from Lan et al.³, resolved from the pixel to the country level, which is an updated version of the NOAA source signature database⁶⁷, wetland estimates from Oh et al.⁶⁸ at $0.5^{\circ} \times 0.5^{\circ}$ grids, and geological seep estimates from Etiope et al.⁶³ at $1^{\circ} \times 1^{\circ}$ grids. Prior estimates of $\delta\text{D-CH}_4$ source signatures include wetland estimates from Stell et al.¹¹ at $0.0833^{\circ} \times 0.0833^{\circ}$ grids, and sector-specific global averages for other sectors from Sherwood et al.⁶⁷. Supplementary Table S1 and Supplementary Fig. S20 provide a summary of the global mean $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ source signatures across these categories.

We compute the ethane:methane emission ratio based on emission estimates of ethane in TZOMPASOSA⁶⁹ and methane fossil fuel emissions as discussed previously. We set a cap of this ratio at 5, based on estimates from Lan et al.³. The prior estimate of ethane emissions from fossil is 6.7 Tg/y, and the global-averaged ethane:methane emission ratio is 0.08. Other ethane emissions include biofuel (2.8 Tg/y⁶⁹), biomass burning (3.5 Tg/y⁷⁰), ship (0.2 Tg/y⁷¹), and aircraft (0.3 Gg/y⁷²).

The primary methane sink is oxidation by hydroxyl radicals (OH) in the troposphere. We use model 3D monthly archived OH from GEOS-Chem v5 benchmark simulations to represent this process⁷³. The prior estimate of methane oxidized by tropospheric OH is 470–480 Tg/y. Additional methane sinks include tropospheric removal by chlorine (Cl, 9–10 Tg/y), based on model 3D monthly Cl from Sherwen et al.⁷⁴, stratospheric removal (38–39 Tg/y), derived from NASA Global Modeling Initiative (GMI) monthly loss frequencies⁷⁵, and soil uptake (34 Tg/y, following Fung et al.⁶⁴).

Methane isotopologues ($^{13}\text{CH}_4$ and CH_3D) are removed via the same processes as $^{12}\text{CH}_4$, but slower due to kinetic isotope effects (KIEs). Supplementary Table S1 summarizes the KIE values used in our model, drawn from prior laboratory and modeling studies^{76–84}. For OH oxidation of $^{13}\text{CH}_4$, reported KIE values^{76–84} range from 1.0039 to 1.0061. To assess model sensitivity to KIE parameters, we conduct simulations using three representative values (1.0039, 1.0046, and 1.0061). As shown in Supplementary

Fig. S16, the best agreement with observational data is achieved using a KIE of 1.0039, which we adopt in our simulations. To further evaluate the validity of this choice, we compare the prior and optimized simulations to observed $\delta^{13}\text{C}$ -CH₄ seasonality in the Southern Hemisphere. The optimized simulation closely reproduces the seasonal cycle (Supplementary Fig. S17), lending additional support to the adopted KIE and the robustness of our inversion results.

Ethane sinks include oxidation by OH, NO₃, Cl, and Br in the troposphere. We simulate the ethane loss based on kinetic data from the International Union of Pure and Applied Chemistry (IUPAC⁸⁵), and 3D monthly fields from GEOS-Chem v14.0.1. Stratospheric ethane loss is computed in the model using GMI loss frequencies⁷⁵.

We derive the model initial conditions (01/01/2018) for methane, $\delta^{13}\text{C}$ -CH₄, and δD -CH₄, in three steps. First, we estimate latitude-averaged initial values for $\delta^{13}\text{C}$ -CH₄ and δD -CH₄ using NOAA observations²¹, aggregated into 4°-latitude bins, and apply methane concentration fields optimized from a global inversion⁵. Second, we conduct a 50-years global model spin-up to generate a dynamically consistent initial state that captures the spatial distribution of methane and its isotopologues. Third, we refine these initial conditions through a 6-month 4D-Var inversion (01–06/2018), assimilating TROPOMI satellite methane retrievals and in situ isotopic measurements. Initial conditions for ethane are derived from a 2-year full-chemistry GEOS-Chem v14.0.1 simulation, consistent with the benchmark setup⁸⁶. To further minimize potential biases associated with initialization, we conduct the full inversion over the period 01/2018–06/2022 and focus our analysis on the period 01/2019–12/2021.

Methane 4D-Var inversion framework

The inversion framework is an updated version of the 4D-Var Bayesian methane optimization in the GEOS-Chem adjoint model^{27, 54, 87}, through iteratively minimizing the following cost function $J(\mathbf{x})$.

$$J(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_a)^T \mathbf{S}_a^{-1} (\mathbf{x} - \mathbf{x}_a) + \gamma (\mathbf{y} - F(\mathbf{x}))^T \mathbf{S}_o^{-1} (\mathbf{y} - F(\mathbf{x})) \quad (1)$$

where $\mathbf{x}$ is the state vector to be optimized, including grid-level monthly methane total emissions, grid-level monthly $\delta^{13}\text{C}$ -CH₄ source signatures, grid-level monthly δD -CH₄ source signatures, and grid-level monthly ethane:methane emission ratios, also optimizing grid-level annual or multiple-year averaged OH concentration in the optimized OH cases; $\mathbf{x}_a$ is the prior estimates of the state vector; $\mathbf{S}_a$ is the prior error covariance, as described later; $\mathbf{y}$ is a vector of observations, as discussed in previous, F is the chemical transport model operator to project emission estimates into concentration simulations; $\mathbf{S}_o$ is the observing system error covariance, as described later; γ is a regularization parameters to balance contributions to the cost function between the term measuring departures from the prior estimates and the term measuring the observation-simulation mismatches.

Our inversions include three treatments of OH: 1) use the prior estimates of OH as a fixed constraint, 2) optimize multi-year-average grid-level OH concentrations, and 3) optimize annual grid-level OH concentrations. The optimized OH concentrations from our inversions are close to their prior estimates, which are from full-chemistry simulation in GEOS-Chem v5 model and have been shown to agree well with aircraft measurements of atmospheric background OH and CO⁵. In the main text, we refer to the ensemble mean of the inverse solutions with these three OH treatments, unless otherwise specified.

The prior error covariance $\mathbf{S}_a$ combines uncertainty estimates for methane emissions at sectoral level in quadrature. This includes the one standard deviation of WetCHARTs model ensemble, the grid-level uncertainty estimates from GFEI v2⁵⁸, and uncertainties computed based on a scale-dependent

approach⁶¹ for regional anthropogenic emissions, including Scarpelli estimates for Canada⁵⁹ and Mexico⁶⁰, and Gridded EPA inventory for US⁶¹. We assume uncertainties of other emission sectors are 50%. The resulting uncertainty estimates of methane total emissions are on average 71% at a grid level. Previous studies^{55, 73, 88} indicate that the spatial covariance of methane emissions has a 200–500 km correlation length. Our model resolution ($4^\circ \times 5^\circ$) is at the top-edge of this length. We thus employ a diagonal prior error covariance matrix. Furthermore, we assume 100% uncertainties in the estimates of isotope signatures of $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$, and ethane:methane emission ratios. To evaluate the influence of prior error covariance on the inversion solutions, we perform a sensitivity test reducing the uncertainty estimates of isotopic source signatures to 50%. This sensitivity test limit large deviations in isotopic source signatures unless they are strongly supported by the data. The results show that the spatial distribution and total emissions remain robust. For the year 2019, the optimized total emissions, global-averaged $\delta^{13}\text{C-CH}_4$ source signature, and global-averaged $\delta\text{D-CH}_4$ source signature, vary within 0.9 Tg/y, 0.02‰, and 0.02‰, respectively. We also assume 50% uncertainties in the grid-level OH concentration estimates, following earlier studies⁵.

Elements in the observing system error covariance $\mathbf{S}_0$ of TROPOMI methane observations include three components, combined in quadrature: instrument precision from the measurement database, model transport error estimated through simulations with varying meteorological fields and model platforms, as detailed in our previous study²⁷, and representation error as one standard deviation of daily measurements at the model spatial resolution. On average, the observing system error of TROPOMI methane is 53.04 ppb². We compute observing system error covariance of $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and ethane in a similar process, including model transport error that is estimated in the same approach as methane, and measurement accuracy from the measurement database. For data lacking accuracy estimates, we use the difference between paired measurements or instrument precision. On average, the observing system errors are 0.2‰, 19‰, and 43 ppt, for $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and ethane, respectively. We assume the observing system error covariance is diagonal.

The regularization parameters γ are defined in two steps. First, we apply regularization parameters separately to each tracer (methane, $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and ethane) for an equivalent contribution from each tracer to the cost function. Second, we perform sensitivity inversions (γ varying from 10^{-6} to 10^3 , L-curve see Supplementary Fig. S21), further scaling the regularization parameters to balance the optimized solutions between prior estimates and observation constraints. The final regularization parameters are 0.01, 330, 12, and 0.05, for methane, $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and ethane, respectively.

The inversion is solved in a quasi-Newtonian routine, following algorithms in previous studies^{5, 27, 54, 55}, with updated inventories and updated model settings as described above.

Assimilation framework of $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and ethane

Here we introduce the new model capability to assimilate $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and ethane, in this 4D-Var framework, as follows.

Step1: Compute adjoint forcing of isotopic signature. Taking $\delta^{13}\text{C-CH}_4$ for example, its adjoint forcing in terms of isotopic signature is given as:

$$\frac{\partial J(\mathbf{x})}{\partial \mathbf{y}_{\delta^{13}\text{CH}_4}} = 2\gamma \mathbf{S}_0^{-1} (\mathbf{y}_{\delta^{13}\text{CH}_4} - F(\mathbf{x})) \quad (2)$$

where $\mathbf{y}_{\delta^{13}\text{CH}_4}$ is the observed atmospheric $\delta^{13}\text{C-CH}_4$ and $J(\mathbf{x})$ is the cost function in Eq. 1.

Step 2: Compute the adjoint forcing in terms of emissions. Atmospheric $\delta^{13}\text{C}-\text{CH}_4$ is defined as Eq. 3.

$$\delta^{13}\text{CH}_4 = \frac{R_{[13\text{CH}_4]:[12\text{CH}_4]_{\text{sample}}}}{R_{[13\text{CH}_4]:[12\text{CH}_4]_{\text{ref}}}} - 1 \quad (3)$$

where R are molar ratios of $^{13}\text{CH}_4$ to $^{12}\text{CH}_4$. The $R_{[13\text{CH}_4]:[12\text{CH}_4]_{\text{ref}}}$ is equal to 0.01123720, based on the standard reference from the International Atomic Energy Agency (IAEA). We transfer the ratios of $^{13}\text{CH}_4$ to $^{12}\text{CH}_4$ into the ratios of $^{13}\text{CH}_4$ to CH_4 (in the unit of mol:mol) as in Eq. 4, and compute its adjoint forcing through Eq. 5, for observation i .

$$R_{[13\text{CH}_4]:[\text{CH}_4]_i} = 1 / \left(\frac{1}{R_{[13\text{CH}_4]:[12\text{CH}_4]_i}} + 1 \right) \quad (4)$$

$$\frac{\partial J(\mathbf{x})}{\partial R_{[13\text{CH}_4]:[\text{CH}_4]_i}} = \frac{\partial J(\mathbf{x})}{\partial y_{\delta^{13}\text{CH}_4,i}} \frac{\partial y_{\delta^{13}\text{CH}_4,i}}{\partial R_{[13\text{CH}_4]:[\text{CH}_4]_i}} \quad (5)$$

We further compute the adjoint forcing of methane mass ($m_{13\text{CH}_4,i}$ and $m_{\text{CH}_4,i}$, in the unit of kg) in Eq. 6 and Eq. 7, where M_{CH_4} and $M_{13\text{CH}_4}$ are the mole mass.

$$\frac{\partial J(\mathbf{x})}{\partial m_{13\text{CH}_4,i}} = \left(\frac{M_{\text{CH}_4}}{M_{13\text{CH}_4}} \right) \left(\frac{1}{m_{\text{CH}_4,i}} \right) \frac{\partial J(\mathbf{x})}{\partial R_{[13\text{CH}_4]:[\text{CH}_4]_i}} \quad (6)$$

$$\frac{\partial J(\mathbf{x})}{\partial m_{\text{CH}_4,i}} = (-1) \left(\frac{M_{\text{CH}_4}}{M_{13\text{CH}_4}} \right) \left(\frac{m_{13\text{CH}_4,i}}{m_{\text{CH}_4,i}^2} \right) \frac{\partial J(\mathbf{x})}{\partial R_{[13\text{CH}_4]:[\text{CH}_4]_i}} \quad (7)$$

The forward model of methane assumes a linear relationship between emissions and atmospheric mass changes. Thus, the sensitivity of the cost function to emissions E_j at model grid cell j , are in Eq. 8 and 9.

$$\frac{\partial J(\mathbf{x})}{\partial E_{13\text{CH}_4,j}} = \frac{\partial J(\mathbf{x})}{\partial m_{13\text{CH}_4,i}} \frac{\partial m_{13\text{CH}_4,i}}{\partial E_{13\text{CH}_4,j}} \quad (8)$$

$$\frac{\partial J(\mathbf{x})}{\partial E_{\text{CH}_4,j}} = \frac{\partial J(\mathbf{x})}{\partial m_{\text{CH}_4,i}} \frac{\partial m_{\text{CH}_4,i}}{\partial E_{\text{CH}_4,j}} \quad (9)$$

Step 3: Compute the sensitivity of the cost function to methane total emissions and isotopic source signatures, for model grid cell j , based on Eq. 10 and Eq. 11.

$$\frac{\partial J(\mathbf{x})}{\partial s_{13\text{CH}_4,j}} = \left(\frac{M_{13\text{CH}_4}}{M_{\text{CH}_4}} \right) R_{[13\text{CH}_4]:[12\text{CH}_4]_{\text{ref}}} E_{\text{CH}_4,j} \frac{1}{(1 + a_j)^2} \frac{\partial J(\mathbf{x})}{\partial E_{13\text{CH}_4,j}} \quad (10)$$

$$\left(\frac{\partial J(\mathbf{x})}{\partial E_{\text{CH}_4,j}} \right)^* = \frac{\partial J(\mathbf{x})}{\partial E_{\text{CH}_4,j}} + \left(\frac{M_{13\text{CH}_4}}{M_{\text{CH}_4}} \right) \left(\frac{a_j}{1 + a_j} \right) \frac{\partial J(\mathbf{x})}{\partial E_{13\text{CH}_4,j}} \quad (11)$$

where $\mathbf{s}$ is a vector of isotopic source signatures and $\mathbf{a}$ is formulated as $R_{[13\text{CH}_4]:[12\text{CH}_4]_{\text{ref}}}(1 + \mathbf{s}_{13\text{CH}_4})$. The superscript asterisk in Eq. 11 indicates that these partial differential equations represent the sensitivity of the cost function to total methane emissions jointly parameterized with the source signatures of methane total emissions, in contrast to Eq. 9, where the derivative is taken jointly with respect to isotopic mass.

Step 4: The processes to assimilate $\delta\text{D}-\text{CH}_4$ follow a similar process as above, but apply

$R_{[\text{CH}_3\text{D}]:[12\text{CH}_4]_{\text{ref}}} = 155.76 \times 10^{-6}$, for the reference material Vienna Standard Mean Ocean Water.

Step 5: The processes to assimilate ethane include two steps. First, compute the sensitivity of the cost function to ethane emissions (unit: kg) based on ethane observations (unit: ppt), following a comparable process as assimilating methane, detailed in our earlier studies^{5, 27, 55}. Second, compute the sensitivity of cost function to methane total emissions and ethane:methane emission ratio, following a comparable process as the step 3 discussed above.

Refine optimized total emissions into sector level

We refine the optimized total emissions into a sectoral level (i.e., fossil fuel, livestock, wastewater/landfill, rice, wetlands, other anthropogenic, and other natural sources), on a grid basis, building on the optimized emissions and the optimized isotopic source signatures. The sum of these sectoral emission estimates remains the same as the optimized total emissions from the 4D-Var inversions. Specifically, we minimize the cost function $S(sf_{sec})$ in Eq. 12 separately for each model grid cell j , under the condition that the total methane emission remains the same as the 4D-Var solution, as in Eq. 13.

$$S(sf_{sec,j}) = (sf_{sec,j} - 1)^2 + \left(\frac{sig_{sec-\delta^{13}C,j}}{sig_{opt-\delta^{13}C,j}} - 1 \right)^2 + \left(\frac{sig_{sec-\delta D,j}}{sig_{opt-\delta D,j}} - 1 \right)^2 \quad (12)$$

$$\text{Condition: } \sum_{sec} sf_{sec,j} E_{sec-CH_4,j} = E_{opt-CH_4,j} \quad (13)$$

where sf_{sec} is scaling factors of the refined sectoral emissions; $sig_{sec-\delta^{13}C}$ and $sig_{sec-\delta D}$ are isotopic source signatures based on sf_{sec} , while $sig_{opt-\delta^{13}C}$ and $sig_{opt-\delta D}$ are optimized source signatures from the 4D-Var inversions, for $\delta^{13}C-CH_4$ and $\delta D-CH_4$, respectively; E_{sec-CH_4} is prior estimates of sectoral methane emissions, and E_{opt-CH_4} is the 4D-Var optimized methane total emission estimates. As methane emissions have a heavy tailed distribution (i.e., emission hotspots contribute to a large fraction of the total emission), we solve this refined optimization on a log scale iteratively, through Gaussian-Newton approach. We apply this refine operator to the base-case inversions E1–E3, which assimilate methane, $\delta^{13}C-CH_4$, and $\delta D-CH_4$.

Data availability

The GEOS-Chem adjoint model is accessible via https://adjoint.colorado.edu:8080/yanko.davila/gcadj_std, and the specific version used for methane data assimilation has been archived at <https://doi.org/10.13020/g5xc-nj81>⁵⁴. Satellite retrievals of methane from TROPOMI are publicly available at https://sentinels.copernicus.eu/data-products/-/asset_publisher/fp37fc19FN8F/content/tropomi-level-2-methane. Observational datasets from NOAA methane isotopologues are publicly available at <https://gml.noaa.gov/data>, and from ObsPack are publicly available at <https://gml.noaa.gov/ccgg/obspack>.

References

1. IPCC, 2021: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 2391 pp. doi:10.1017/9781009157896.
2. Saunio, M., Martinez, A., Poulter, B., Zhang, Z., Raymond, P., Regnier, P., Canadell, J. G., Jackson, R. B., Patra, P. K., Bousquet, P., Ciais, P., Dlugokencky, E. J., Lan, X., Allen, G. H., Bastviken, D., Beerling, D. J., Belikov, D. A., Blake, D. R., Castaldi, S., Crippa, M., Deemer, B. R., Dennison, F., Etiope, G., Gedney, N., Höglund-Isaksson, L., Holgersson, M. A., Hopcroft, P. O., Hugelius, G., Ito, A., Jain, A. K., Janardanan, R., Johnson, M. S., Kleinen, T., Krummel, P., Lauerwald, R., Li, T., Liu, X., McDonald, K. C., Melton, J. R., Mühle, J., Müller, J., Murguía-Flores, F., Niwa, Y., Noce, S., Pan, S., Parker, R. J., Peng, C., Ramonet, M., Riley, W. J., Rocher-Ros, G., Rosentreter, J. A., Sasakawa, M., Segers, A., Smith, S. J., Stanley, E. H., Thanwerdas, J., Tian, H., Tsuruta, A., Tubiello, F. N., Weber T. S., van der Werf G., Worthy, D. E., Xi, Y., Yoshida, Y., Zhang, W., Zheng, B., Zhu, Q., Zhu, Q., Zhuang, Q., Global methane budget 2000-2020. *Earth System Science Data*, 2025, 17, 1873–1958.
3. Lan, X., Basu, S., Schwietzke, S., Bruhwiler, L.M., Dlugokencky, E.J., Michel, S.E., Sherwood, O.A., Tans, P.P., Thoning, K., Etiope, G., Zhuang, Q., Improved constraints on global methane emissions and sinks using $\delta^{13}\text{C}\text{-CH}_4$. *Global Biogeochemical Cycles*, 2021. 35: e2021GB007000.
4. Lan, X., Thoning, K. W., Dlugokencky, E. J., Trends in globally-averaged CH_4 , N_2O , and SF_6 determined from NOAA Global Monitoring Laboratory measurements. Version 2025-07. Available at: <https://doi.org/10.15138/P8XG-AA10>.
5. Yu, X., Millet, D. B., Henze, D. K., Turner, A. J., Delgado, A. L., Bloom, A. A., Sheng, J., A high-resolution satellite-based map of global methane emissions reveals missing wetland, fossil fuel, and monsoon sources. *Atmospheric Chemistry and Physics*, 2023. 23(5): p. 3325-3346.
6. Feng, L., Palmer, P.I., Parker, R.J., Lunt, M.F. and Bösch, H., Methane emissions are predominantly responsible for record-breaking atmospheric methane growth rates in 2020 and 2021. *Atmospheric Chemistry and Physics*, 2023. 23(8), pp.4863-4880.
7. Stevenson, D. S., Derwent, R. G., Wild, O., Collins, W. J., COVID-19 lockdown emission reductions have the potential to explain over half of the coincident increase in global atmospheric methane. *Atmospheric Chemistry and Physics*, 2022. 22(21): p. 14243-14252.
8. Peng, S., Lin, X., Thompson, R. L., Xi, Y., Liu, G., Hauglustaine, D., Lan, X., Poulter, B., Ramonet, M., Saunio, M., Wetland emission and atmospheric sink changes explain methane growth in 2020. *Nature*, 2022. 612(7940): p. 477-482.
9. Hachmeister, J., Schneising, O., Buchwitz, M., Burrows, J.P., Notholt, J., Buschmann, M., Zonal variability of methane trends derived from satellite data. *Atmospheric Chemistry and Physics*, 2024. 24(1), pp.577-595.
10. Zhang, Z., Poulter, B., Feldman, A.F., Ying, Q., Ciais, P., Peng, S., Li, X., Recent intensification of wetland methane feedback. *Nature Climate Change*, 2023. 13(5), pp.430-433.
11. Stell, A.C., Douglas, P.M., Rigby, M., Ganesan, A.L., The impact of spatially varying wetland source signatures on the atmospheric variability of $\delta\text{D}\text{-CH}_4$. *Philosophical Transactions of the Royal Society A*, 2021, 379(2210), p.20200442.

12. Douglas, P.M., Stratigopoulos, E., Park, J., Phan, D., Geographic variability in freshwater methane hydrogen isotope ratios and its implications for global isotopic source signatures, *Biogeosciences*, 2021. 18, 3505-3527.
13. Michel, S. E., Lan, X., Miller, J., Tans, P., Clark, J. R., Schaefer, H., Sperlich, P., Brailsford, G., Morimoto, S., Moossen, H., Li, J., Rapid shift in methane carbon isotopes suggests microbial emissions drove record high atmospheric methane growth in 2020–2022. *Proceedings of the National Academy of Sciences* 2024, 121 (44), e2411212121.
14. Yin, Y., Chevallier, F., Ciais, P., Bousquet, P., Saunois, M., Zheng, B., Worden, J., Bloom, A. A., Parker, R. J., Jacob, D. J., Dlugokencky, E. J., Frankenberg, C., Accelerating methane growth rate from 2010 to 2017: leading contributions from the tropics and East Asia. *Atmospheric Chemistry and Physics*, 2021, 21 (16), 12631-12647.
15. Jackson, R. B., Saunois, M., Bousquet, P., Canadell, J. G., Poulter, B., Stavert, A. R., Bergamaschi, P., Niwa, Y., Segers, A., Tsuruta, A., Increasing anthropogenic methane emissions arise equally from agricultural and fossil fuel sources. *Environmental Research Letters* 2020, 15 (7), 071002.
16. Zhang, Y., Jacob, D. J., Lu, X., Maasakkers, J. D., Scarpelli, T. R., Sheng, J. X., Shen, L., Qu, Z., Sulprizio, M. P., Chang, J., Bloom, A. A., Ma, S., Worden, J., Parker, R. J., Boesch, H., Attribution of the accelerating increase in atmospheric methane during 2010–2018 by inverse analysis of GOSAT observations. *Atmospheric Chemistry and Physics*, 2021, 21 (5), 3643-3666.
17. Janardanan, R., Maksyutov, S., Wang, F., Nayagam, L., Sahu, S. K., Mangaraj, P., Saunois, M., Lan, X., Matsunaga, T., Country-level methane emissions and their sectoral trends during 2009–2020 estimated by high-resolution inversion of GOSAT and surface observations. *Environmental Research Letters* 2024, 19 (3), 034007.
18. Chandra, N., Patra, P.K., Fujita, R., Höglund-Isaksson, L., Umezawa, T., Goto, D., Morimoto, S., Vaughn, B.H., Röckmann, T., Methane emissions decreased in fossil fuel exploitation and sustainably increased in microbial source sectors during 1990–2020. *Communications Earth & Environment*, 2024. 5(1), p.147.
19. McNorton, J., Wilson, C., Gloor, M., Parker, R.J., Boesch, H., Feng, W., Hossaini, R., Chipperfield, M.P., Attribution of recent increases in atmospheric methane through 3-D inverse modelling. *Atmospheric Chemistry and Physics*, 2018. 18(24), pp.18149-18168.
20. Thanwerdas, J., Saunois, M., Berchet, A., Pison, I., Bousquet, P., Investigation of the renewed methane growth post-2007 with high-resolution 3-D variational inverse modeling and isotopic constraints. *Atmospheric Chemistry and Physics*, 2024. 24(4), pp.2129-2167.
21. Basu, S., Lan, X., Dlugokencky, E., Michel, S., Schwietzke, S., Miller, J.B., Bruhwiler, L., Oh, Y., Tans, P.P., Apadula, F., Gatti, L.V., Estimating emissions of methane consistent with atmospheric measurements of methane and $\delta^{13}\text{C}$ of methane. *Atmospheric Chemistry and Physics*, 2022. 22(23), pp.15351-15377.
22. Worden, J. R., Bloom, A. A., Pandey, S., Jiang, Z., Worden, H. M., Walker, T. W., Houweling, S., Röckmann, T., Reduced biomass burning emissions reconcile conflicting estimates of the post-2006 atmospheric methane budget. *Nature Communications* 2017, 8 (1), 2227.
23. He, J., Naik, V., Horowitz, L. W., Interpreting changes in global methane budget in a chemistry-climate model constrained with methane and isotopic observations. *AGU Advances* 2026, 7 (1), e2025AV001822.
24. Turner, A. J., Frankenberg, C., Wennberg, P. O., Jacob, D. J., Ambiguity in the causes for decadal trends in atmospheric methane and hydroxyl. *Proceedings of the National Academy of Sciences*, 2017. 114(21): p. 5367-5372.
25. Rigby, M., Montzka, S. A., Prinn, R. G., White, J. W. C., Young, D., O'Doherty, S., Lunt, M. F., Ganesan, A. L., Manning, A. J., Simmonds, P. G., Role of atmospheric oxidation in recent

- methane growth. *Proceedings of the National Academy of Sciences*, 2017. 114(21): p. 5373-5377.
26. van Herpen, M. M., Li, Q., Saiz-Lopez, A., Liisberg, J. B., Röckmann, T., Cuevas, C. A., Fernandez, R. P., Mak, J. E., Mahowald, N. M., Hess, P., Meidan, D., Photocatalytic chlorine atom production on mineral dust-sea spray aerosols over the North Atlantic. *Proceedings of the National Academy of Sciences*, 2023. 120, e2303974120.
 27. Yu, X., D.B. Millet, Henze, D.K., How well can inverse analyses of high-resolution satellite data resolve heterogeneous methane fluxes? Observing system simulation experiments with the GEOS-Chem adjoint model (v35). *Geoscientific Model Development*, 2021. 14(12): p. 7775-7793.
 28. Jacob, D. J., Varon, D. J., Cusworth, D. H., Dennison, P. E., Frankenberg, C., Gautam, R., Guanter, L., Kelley, J., McKeever, J., Ott, L. E., Quantifying methane emissions from the global scale down to point sources using satellite observations of atmospheric methane. *Atmospheric Chemistry and Physics*, 2022. 22(14): p. 9617-9646.
 29. Rice, A.L., Butenhoff, C.L., Teama, D.G., Röger, F.H., Khalil, M.A.K., Rasmussen, R.A., Atmospheric methane isotopic record favors fossil sources flat in 1980s and 1990s with recent increase. *Proceedings of the National Academy of Sciences*, 2016. 113(39), pp.10791-10796.
 30. Schwietzke, S., Sherwood, O.A., Bruhwiler, L.M., Miller, J.B., Etiope, G., Dlugokencky, E.J., Michel, S.E., Arling, V.A., Vaughn, B.H., White, J.W., Tans, P.P., Upward revision of global fossil fuel methane emissions based on isotope database. *Nature*, 2016. 538(7623), pp.88-91.
 31. Lan, X., Tans, P., Sweeney, C., Andrews, A., Dlugokencky, E., Schwietzke, S., Kofler, J., McKain, K., Thoning, K., Crotwell, M., Montzka, S., Long-term measurements show little evidence for large increases in total US methane emissions over the past decade. *Geophysical Research Letters*, 2019. 46(9), pp.4991-4999.
 32. Pendergrass, D. C., Jacob, D. J., Balasus, N., Estrada, L., Varon, D. J., East, J. D., He, M., Mooring, T. A., Penn, E., Nesser, H., Worden, J. R., Trends and seasonality of 2019–2023 global methane emissions inferred from a localized ensemble transform Kalman filter (CHEEREIO v1.3.1) applied to TROPOMI satellite observations. *Atmospheric Chemistry and Physics*, 2025, 25 (21), 14353-14369.
 33. Hmiel, B., Petrenko, V.V., Dyonisius, M.N., Buizert, C., Smith, A.M., Place, P.F., Harth, C., Beaudette, R., Hua, Q., Yang, B., Vimont, I., Preindustrial $^{14}\text{CH}_4$ indicates greater anthropogenic fossil CH_4 emissions. *Nature*, 2020. 578(7795), pp.409-412.
 34. Turner, A.J., C. Frankenberg, Kort, E. A., Interpreting contemporary trends in atmospheric methane. *Proceedings of the National Academy of Sciences*, 2019. 116(8): p. 2805-2813.
 35. Ciais, P., Zhu, Y., Cai, Y., Lan, X., Michel, S. E., Zheng, B., Zhao, Y., Hauglustaine, D. A., Lin, X., Zhang, Y., Sun, S., Tian, X., Zhao, M., Wang, Y., Chang, J., Dou, X., Liu, Z., Andrew, R., Quinn, C. A., Poulter, B., Ouyang, Z., Yuan, W., Yuan, K., Zhu, Q., Li, F., Pan, N., Tian, H., Yu, X., Rocher-Ros, G., Johnson, M. S., Li, M., Li, M., Feng, D., Raymond, P., Yang, X., Canadell, J. G., Jackson, R. B., Yu, X., Li, Y., Saunois, M., Bousquet P., Peng, S., Why methane surged in the atmosphere during the early 2020s. *Science* 2026, 391 (6785), eadx8262.
 36. Chen, Z., Balasus, N., Lin, H., Nesser, H., Jacob, D. J., African rice cultivation linked to rising methane. *Nature Climate Change* 2024, 14 (2), 148-151.
 37. Dowd, E., Wilson, C., Chipperfield, M. P., Gloor, E., Manning, A., Doherty, R., Decreasing seasonal cycle amplitude of methane in the northern high latitudes being driven by lower-latitude changes in emissions and transport. *Atmospheric Chemistry and Physics*, 2023, 23 (13), 7363-7382.

38. Liu, G., Shen, L., Ciais, P., Lin, X., Hauglustaine, D., Lan, X., Turner, A. J., Xi, Y., Zhu, Y., Peng, S., Trends in the seasonal amplitude of atmospheric methane. *Nature* 2025, 641 (8063), 660-665.
39. Zhang, Y., Fang, S., Chen, J., Lin, Y., Chen, Y., Liang, R., Jiang, K., Parker, R. J., Boesch, H., Steinbacher, M., Sheng, J. X., Lu, X., Song, S., Peng, S., Observed changes in China's methane emissions linked to policy drivers. *Proceedings of the National Academy of Sciences* 2022, 119 (41), e2202742119.
40. Qian, H., Zhu, X., Huang, S., Linquist, B., Kuzyakov, Y., Wassmann, R., Minamikawa, K., Martinez-Eixarch, M., Yan, X., Zhou, F., Sander, B.O., Greenhouse gas emissions and mitigation in rice agriculture, *Nature Reviews Earth & Environment*, 2023. 4(10), pp.716-732.
41. East, J. D., Jacob, D. J., Balasus, N., Bloom, A. A., Bruhwiler, L., Chen, Z., Kaplan, J. O., Mickley, L. J., Mooring, T. A., Penn, E., Poulter, B., Sulprizio, M. P., Worden, J. R., Yantosca, R. M., Zhang, Z., Interpreting the seasonality of atmospheric methane. *Geophysical Research Letters* 2024, 51 (10), e2024GL108494.
42. Ma, S., Worden, J.R., Bloom, A.A., Zhang, Y., Poulter, B., Cusworth, D.H., Yin, Y., Pandey, S., Maasackers, J.D., Lu, X., Shen, L., Satellite constraints on the latitudinal distribution and temperature sensitivity of wetland methane emissions. *AGU Advances*, 2021. 2(3), p.e2021AV000408.
43. Colosimo, F., Thomas, R., Lloyd, J.R., Taylor, K.G., Boothman, C., Smith, A.D., Lord, R., Kalin, R.M., Biogenic methane in shale gas and coal bed methane: A review of current knowledge and gaps. *International Journal of Coal Geology*, 2016. 165, pp.106-120.
44. Lorente, A., Borsdorff, T., Butz, A., Hasekamp, O., aan de Brugh, J., Schneider, A., Wu, L., Hase, F., Kivi, R., Wunch, D., Methane retrieved from TROPOMI: improvement of the data product and validation of the first 2 years of measurements. *Atmospheric Measurement Techniques*, 2021. 14(1): p. 665-684.
45. Lorente, A., Borsdorff, T., Martinez-Velarte, M.C., Landgraf, J., Accounting for surface reflectance spectral features in TROPOMI methane retrievals. *Atmospheric Measurement Techniques Discussions*, 2022, pp.1-15.
46. Lan Xi, Dlugokencky, E., Michel, S. E., Basu, S., Schuldt, K., Mund, J., Aoki, S., di Sarra, A., Vermeulen, A., Andrews, A., Jordan, A., Baier, B., Labuschagne, C., Myhre, C. L., Sweeney, C., Kubistin, D., Smale, D., Worthy, D., Cuevas, E., Apadula, F., Brailsford, G., Lee, H., Moosen, H., Schaefer, H., Jui, H., Necki, J., Arduini, J., Miller, J., Moncrieff, J., Hatakka, J., Uhse, K., McKain, K., Haszpra, L., Gatti, L., Ries, L., Steinbacher, M., Schmidt, M., Ramonet, M., Arshinov, M., Sasakawa, M., Paramonova, N., Bergamaschi, P., Langenfelds, R., Kim, S., Morimoto, S., Takatsuji, S., Nichol, S., Umezawa, T., Di Iorio, T., Kawasaki, T., Database of methane (CH₄) abundance and its stable carbon isotope (d¹³CH₄) composition from atmospheric measurements, 2022, <https://doi.org/10.15138/64w0-0g71>.
47. The World Data Centre for Greenhouse Gases (WDCGG) data archive. <https://gaw.kishou.go.jp>.
48. Umezawa, T., Brenninkmeijer, C.A., Röckmann, T., Van Der Veen, C., Tyler, S.C., Fujita, R., Morimoto, S., Aoki, S., Sowers, T., Schmitt, J., Bock, M., Interlaboratory comparison of δ¹³C and δD measurements of atmospheric CH₄ for combined use of data sets from different laboratories. *Atmospheric Measurement Techniques*, 2018. 11(2), pp.1207-1231.
49. Lan, X., Dlugokencky, E., Michel, S. E., Basu, S., Schuldt, K., Mund, J., Aoki, S., Sarra, A. G., Vermeulen, A., Andrews, A., ... Kawasaki, T., Database of methane (CH₄) abundance and its stable carbon isotope (d¹³CH₄) composition from atmospheric measurements. <https://doi.org/10.15138/64w0-0g71>, 2022.
50. Menoud, M., van Der Veen, C., Scheeren, B., Chen, H., Szénási, B., Morales, R.P., Pison, I., Bousquet, P., Brunner, D., Röckmann, T., Characterisation of methane sources in Lutjewad, The

- Netherlands, using quasi-continuous isotopic composition measurements. *Tellus B: Chemical and Physical Meteorology*, 2020. 72(1), pp.1-20.
51. Röckmann, T., Eyer, S., Van Der Veen, C., Popa, M.E., Tuzson, B., Monteil, G., Houweling, S., Harris, E., Brunner, D., Fischer, H., Zazzeri, G., In situ observations of the isotopic composition of methane at the Cabauw tall tower site. *Atmospheric Chemistry and Physics*, 2016. 16(16), pp.10469-10487.
 52. Davis, K. J., Obland, M. D., Lin, B., Lauvaux, T., O'Dell, C., Meadows, B., Browell, E. V., Crawford, J. H., Digangi, J. P., Sweeney, C., McGill, M. J., Dobler, J., Barrick, J. D., Nehrir, A. R.: ACT-America: L3 Merged In Situ Atmospheric Trace Gases and Flask Data, Eastern USA, ORNL Distributed Active Archive Center, 2018. <https://doi.org/10.3334/ORNLDAAAC/1593>.
 53. Schuldt, K. N., Mund, J., Aalto, T., Arlyn Andrews, Apadula, F., Jgor Arduini, Arnold, S., Baier, B., Bani, L., Bartyzel, J., Bergamaschi, P., Biermann, T., Biraud, S. C., Pierre-Eric Blanc, Boenisch, H., Brailsford, G., Brand, W. A., Brunner, D., Bui, T. P. V., ... Mirosław Zimnoch, Multi-laboratory compilation of atmospheric carbon dioxide data for the period 1983-2022; obspack_ch4_1_GLOBALVIEWplus_v6.0_2023-12-01 [Dataset]. NOAA Global Monitoring Laboratory, 2023. <https://doi.org/10.25925/20231001>.
 54. Yu, X., D.B. Millet, Henze, D. K., Code Updates of GEOS-Chem Adjoint v35 for TROPOMI Methane 4D-Var Inversion. Data Repository for the University of Minnesota. 2021. <https://doi.org/10.13020/g5xc-nj81>.
 55. Yu, X., Millet, D. B., Wells, K. C., Henze, D. K., Cao, H., Griffis, T. J., Kort, E. A., Plant, G., Deventer, M. J., Kolka, R. K., Aircraft-based inversions quantify the importance of wetlands and livestock for Upper Midwest methane emissions. *Atmospheric Chemistry and Physics*, 2021. 21(2): p. 951-971.
 56. Global Modeling and Assimilation Office (GMAO). GEOS Near-Real Time Data Products. https://gmao.gsfc.nasa.gov/GMAO_products/NRT_products.php.
 57. Emission Database for Global Atmospheric Research (EDGAR). <https://edgar.jrc.ec.europa.eu>.
 58. Scarpelli, T.R., Jacob, D.J., Grossman, S., Lu, X., Qu, Z., Sulprizio, M.P., Zhang, Y., Reuland, F., Gordon, D., Worden, J.R., Updated Global Fuel Exploitation Inventory (GFEI) for methane emissions from the oil, gas, and coal sectors: evaluation with inversions of atmospheric methane observations, *Atmospheric Chemistry and Physics*, 2022. 22(5), pp.3235-3249.
 59. Scarpelli, T.R., Jacob, D.J., Moran, M., Reuland, F. and Gordon, D., A gridded inventory of Canada's anthropogenic methane emissions, *Environmental Research Letters*, 2022. 17(1), p.014007.
 60. Scarpelli, T.R., Jacob, D.J., Villasana, C.A.O., Hernández, I.F.R., Moreno, P.R.C., Alfaro, E.A.C., García, M.Á.G., Zavala-Araiza, D., A gridded inventory of anthropogenic methane emissions from Mexico based on Mexico's national inventory of greenhouse gases and compounds. *Environmental Research Letters*, 2020. 15(10), p.105015.
 61. Maasakkers, J. D., Jacob, D. J., Sulprizio, M. P., Turner, A. J., Weitz, M., Wirth, T., Hight, C., DeFigueiredo, M., Desai, M., Schmeltz, R., Gridded national inventory of U.S. methane emissions. *Environmental Science & Technology*, 2016. 50(23): p. 13123-13133.
 62. Bloom, A. A., Bowman, K. W., Lee, M., Turner, A. J., Schroeder, R., Worden, J. R., Weidner, R., McDonald, K. C., Jacob, D. J., A global wetland methane emissions and uncertainty dataset for atmospheric chemical transport models (WetCHARTs version 1.0). *Geoscientific Model Development*, 2017, 10 (6), 2141-2156.
 63. Etiope, G., Ciotoli, G., Schwietzke, S., Schoell, M., Gridded maps of geological methane emissions and their isotopic signature. *Earth System Science Data*, 2019. 11(1), pp.1-22.

64. Fung, I., John, J., Lerner, J., Matthews, E., Prather, M., Steele, L. P., Fraser, P. J., Three-dimensional model synthesis of the global methane cycle. *Journal of Geophysical Research: Atmospheres*, 1991. 96(D7): p. 13033-13065.
65. Darmenov, A.S., Silva, A.d., The Quick Fire Emissions Dataset (QFED): Documentation of versions 2.1, 2.2 and 2.4. 2015: NASA Technical Report Series on Global Modeling and Data Assimilation NASA TM-2015-104606. <https://ntrs.nasa.gov/citations/20180005253>.
66. Doumbia, T., Granier, C., Elguindi, N., Bouarar, I., Darras, S., Brasseur, G., Gaubert, B., Liu, Y., Shi, X., Stavrakou, T., Changes in global air pollutant emissions during the COVID-19 pandemic: a dataset for atmospheric modeling. *Earth System. Science Data*, 2021. 13(8): p. 4191-4206.
67. Sherwood, O.A., Schwietzke, S., Arling, V.A. and Etiope, G., Global inventory of gas geochemistry data from fossil fuel, microbial and burning sources, version 2017. *Earth System Science Data*, 2017. 9(2), pp.639-656.
68. Oh, Y., Zhuang, Q., Welp, L.R., Liu, L., Lan, X., Basu, S., Dlugokencky, E.J., Bruhwiler, L., Miller, J.B., Michel, S.E., Schwietzke, S., Improved global wetland carbon isotopic signatures support post-2006 microbial methane emission increase. *Communications Earth & Environment*, 2022. 3(1), p.159.
69. Tzompa-Sosa, Z.A., Mahieu, E., Franco, B., Keller, C.A., Turner, A.J., Helmig, D., Fried, A., Richter, D., Weibring, P., Walega, J., Yacovitch, T.I., Revisiting global fossil fuel and biofuel emissions of ethane. *Journal of Geophysical Research: Atmospheres*, 2017. 122(4), pp.2493-2512.
70. Global Fire Emissions Database (GFED). <https://www.globalfiredata.org>.
71. Hoesly, R.M., Smith, S.J., Feng, L., Klimont, Z., Janssens-Maenhout, G., Pitkanen, T., Seibert, J.J., Vu, L., Andres, R.J., Bolt, R.M., Bond, T.C., Historical (1750–2014) anthropogenic emissions of reactive gases and aerosols from the Community Emissions Data System (CEDS). *Geoscientific Model Development*, 2018. 11(1), pp.369-408.
72. Simone, N.W., Stettler, M.E., Barrett, S.R., Rapid estimation of global civil aviation emissions with uncertainty quantification. *Transportation Research Part D: Transport and Environment*, 2013. 25, pp.33-41.
73. Wecht, K.J., Jacob, D.J., Frankenberg, C., Jiang, Z., Blake, D.R., Mapping of North American methane emissions with high spatial resolution by inversion of SCIAMACHY satellite data. *Journal of Geophysical Research: Atmospheres*, 2014. 119(12), pp.7741-7756.
74. Sherwen, T., Schmidt, J. A., Evans, M. J., Carpenter, L. J., Großmann, K., Eastham, S. D., Jacob, D. J., Dix, B., Koenig, T. K., Sinreich, R., Global impacts of tropospheric halogens (Cl, Br, I) on oxidants and composition in GEOS-Chem. *Atmospheric Chemistry and Physics*, 2016. 16(18): p. 12239-12271.
75. Murray, L.T., J.A. Logan, Jacob, D.J., Interannual variability in tropical tropospheric ozone and OH: the role of lightning. *Journal of Geophysical Research: Atmospheres*, 2013. 118(19): p. 11,468-11,480.
76. Saueressig, G., Crowley, J.N., Bergamaschi, P., Brühl, C., Brenninkmeijer, C.A., Fischer, H., Carbon 13 and D kinetic isotope effects in the reactions of CH₄ with O (1D) and OH: new laboratory measurements and their implications for the isotopic composition of stratospheric methane. *Journal of Geophysical Research: Atmospheres*, 2001. 106(D19), pp.23127-23138.
77. Saueressig, G., Bergamaschi, P., Crowley, J.N., Fischer, H., and Harris, G.W., Carbon kinetic isotope effect in the reaction of CH₄ with Cl atoms. *Geophysical Research Letters*, 1995. 22(10), pp.1225-1228.
78. Saueressig, G., Bergamaschi, P., Crowley, J.N., Fischer, H., Harris, G.W., D/H kinetic isotope effect in the reaction CH₄+Cl. *Geophysical Research Letters*, 1996. 23(24), pp.3619-3622.

79. Snover, A.K., Quay, P.D., Hydrogen and carbon kinetic isotope effects during soil uptake of atmospheric methane. *Global Biogeochemical Cycles*, 2000. 14(1), pp.25-39.
80. Snover, A.K., Quay, P.D., Hao, W.M., The D/H content of methane emitted from biomass burning. *Global biogeochemical cycles*, 2000. 14(1), pp.11-24.
81. Tyler, S.C., Crill, P.M., Brailsford, G.W., $^{13}\text{C}/^{12}\text{C}$ Fractionation of methane during oxidation in a temperate forested soil. *Geochimica et Cosmochimica Acta*, 1994. 58(6), pp.1625-1633.
82. Reeburgh, W.S., Hirsch, A.I., Sansone, F.J., Popp, B.N., Rust, T.M., Carbon kinetic isotope effect accompanying microbial oxidation of methane in boreal forest soils. *Geochimica et Cosmochimica Acta*, 1997. 61(22), pp.4761-4767.
83. King, S.L., Quay, P.D., Lansdown, J.M., The $^{13}\text{C}/^{12}\text{C}$ kinetic isotope effect for soil oxidation of methane at ambient atmospheric concentrations. *Journal of Geophysical Research: Atmospheres*, 1989. 94(D15), pp.18273-18277.
84. Gierczak, T., Talukdar, R.K., Herndon, S.C., Vaghjiani, G.L., Ravishankara, A.R., Rate coefficients for the reactions of hydroxyl radicals with methane and deuterated methanes. *The Journal of Physical Chemistry A*, 1997. 101(17), pp.3125-3134.
85. IUPAC Task Group on Atmospheric Chemical Kinetic Data Evaluation I. <https://iupac-aeris.ipsl.fr>.
86. GEOS-Chem 14.0.0. https://wiki.seas.harvard.edu/geos-chem/index.php/GEOS-Chem_14.0.0.
87. Henze, D.K., A. Hakami, Seinfeld, J.H., Development of the adjoint of GEOS-Chem. *Atmospheric Chemistry and Physics*, 2007. 7(9): p. 2413-2433.
88. Monteil, G., Houweling, S., Butz, A., Guerlet, S., Schepers, D., Hasekamp, O., Frankenberg, C., Scheepmaker, R., Aben, I., Röckmann, T.. Comparison of CH_4 inversions based on 15 months of GOSAT and SCIAMACHY observations. *Journal of Geophysical Research: Atmospheres*, 2013. 118(20), pp.11-807.

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Competing interest declaration

The authors declare no competing interests.

Tables

Table 1. Inversion setups and performance

Inversion	Assimilated Tracers	OH Treatments	Simulation Bias (ppb) ObsPack	Simulation Bias (ppb) TROPOMI
Prior			-89	-71
A1	CH ₄	Fixed OH	-9	0
A2	CH ₄	Optimized multi-year OH	-10	0
B1	CH ₄ + $\delta^{13}\text{C}$ -CH ₄	Fixed OH	-11	-2
B2	CH ₄ + $\delta^{13}\text{C}$ -CH ₄	Optimized multi-year OH	-8	0
C1	CH ₄ + δD -CH ₄	Fixed OH	-15	1
C2	CH ₄ + δD -CH ₄	Optimized multi-year OH	-14	1
D1	CH ₄ + C ₂ H ₆	Fixed OH	-10	0
D2	CH ₄ + C ₂ H ₆	Optimized multi-year OH	-10	1
E1	CH ₄ + $\delta^{13}\text{C}$ -CH ₄ + δD -CH ₄	Fixed OH	0	7
E2	CH ₄ + $\delta^{13}\text{C}$ -CH ₄ + δD -CH ₄	Optimized multi-year OH	-2	6
E3	CH ₄ + $\delta^{13}\text{C}$ -CH ₄ + δD -CH ₄	Optimized annual OH	-2	6
F1	CH ₄ + $\delta^{13}\text{C}$ -CH ₄ + δD -CH ₄ + C ₂ H ₆	Fixed OH	-12	-2
F2	CH ₄ + $\delta^{13}\text{C}$ -CH ₄ + δD -CH ₄ + C ₂ H ₆	Optimized multi-year OH	-8	2
F3	CH ₄ + $\delta^{13}\text{C}$ -CH ₄ + δD -CH ₄ + C ₂ H ₆	Optimized annual OH	-11	-1

Table 2. The 2019–2021 global and tropical methane budgets

	Total emissions (Tg/y)			Anthropogenic Emissions (Tg/y)			Nature Emissions (Tg/y)			Source Signature ¹ (‰)			
	Prior ²	Optimized ^{2,4}	Isotopic-informed Optimized ^{2,5,6}	Prior ²	Optimized ^{2,4}	Isotopic-informed Optimized ^{2,5,6}	Prior ²	Optimized ^{2,4}	Isotopic-informed Optimized ^{2,5,6}	$\delta^{13}\text{C-CH}_4$ prior ³	$\delta^{13}\text{C-CH}_4$ optimized ^{3,5}	$\delta\text{D-CH}_4$ prior ²	$\delta\text{D-CH}_4$ optimized ^{2,5}
Global	517	602	623 [585–643]	336	368	403 [362–407]	181	234	220 [213–246]	-55.96	-53.74	-288.60	-261.27
East Asia	64	65	91 [62–92]	60	60	87 [59–88]	3	4	4 [4–4]	-51.33	-49.24	-272.46	-227.21
South Asia	43	49	56 [48–57]	41	46	53 [45–53]	3	4	3 [3–4]	-59.45	-53.72	-295.52	-261.08
Southeast Asia	31	42	41 [38–44]	15	19	18 [18–19]	16	24	23 [20–26]	-54.22	-53.39	-295.79	-222.84
Central Africa	47	55	61 [52–65]	17	18	21 [17–21]	30	37	40 [34–45]	-55.45	-49.11	-276.50	-237.11
Brazil	51	66	61 [59–72]	19	21	21 [20–23]	32	45	40 [38–49]	-60.32	-56.59	-300.87	-263.70
Northern Latin America	23	30	26 [25–30]	12	14	13 [13–15]	11	15	13 [12–16]	-57.70	-54.93	-295.10	-277.94

¹Source signature of methane total emissions. Isotopic source signatures by sectors are provided in Supplementary Table S1.²Average of 2019–2021³Year 2019⁴Optimized emissions from methane-only inversions, ensemble mean of fixed OH and optimized OH cases (A1–A2)⁵Optimized emissions or source signatures from methane-isotopic joint inversions, ensemble mean of fixed OH and optimized OH cases (E1–E3)⁶Uncertainty estimates are the range of the inversion ensemble (see Table 1)

Figure Legends

Figure 1. Methane total column increases.

Increases of atmospheric total column methane, from 2018 to 2021, based on TROPOMI satellite measurements, oversampled to $0.1^\circ \times 0.1^\circ$.

Figure 2. Methane emission corrections after implementing isotopic constraints.

Emission corrections after implementing isotopic constraints, compared with the methane-only inversions. The color shading (blue to red) shows the average emission corrections for 2019–2021, calculated as the solution from the methane-isotopic joint inversion minus the methane-only solution. The bar charts present emission corrections by sectors in regional budgets (wetlands: white; fossil: light purple; agriculture: medium purple; wastewater: heavy purple; other sources: gray). Downward emission corrections are shown with striped bars, and upward corrections with solid bars. Regional boundaries are outlined and labeled in the lower-left panel. Sectoral emission budgets are shown in Supplementary Fig. 10.

Figure 3. The optimized methane emissions after implementing isotopic constraints.

Optimized methane emission distributions from the methane-isotopic joint inversion. The bar charts present regional methane emission budgets by sectors, for Brazil, Northern Latin America, Central Africa, South Asia, East Asia, and Southeast Asia. Regional boundaries are outlined and labeled in the lower-left panel of Figure 2.

Figure 4. Emission corrections to the prior inventories.

Emission corrections to the prior bottom-up inventories. The color shading (blue to red) shows the average emission corrections for 2019–2021, calculated as the solution from the methane-isotopic joint inversion minus the bottom-up prior estimates. The bar charts illustrate emission corrections by sectors for the underestimated hotspots, region boundaries indicated by the squares. Monthly emission corrections are shown in Supplementary Fig. S22–S24.

Figure 5. Time series of methane column concentration, global emissions, and emission underestimates.

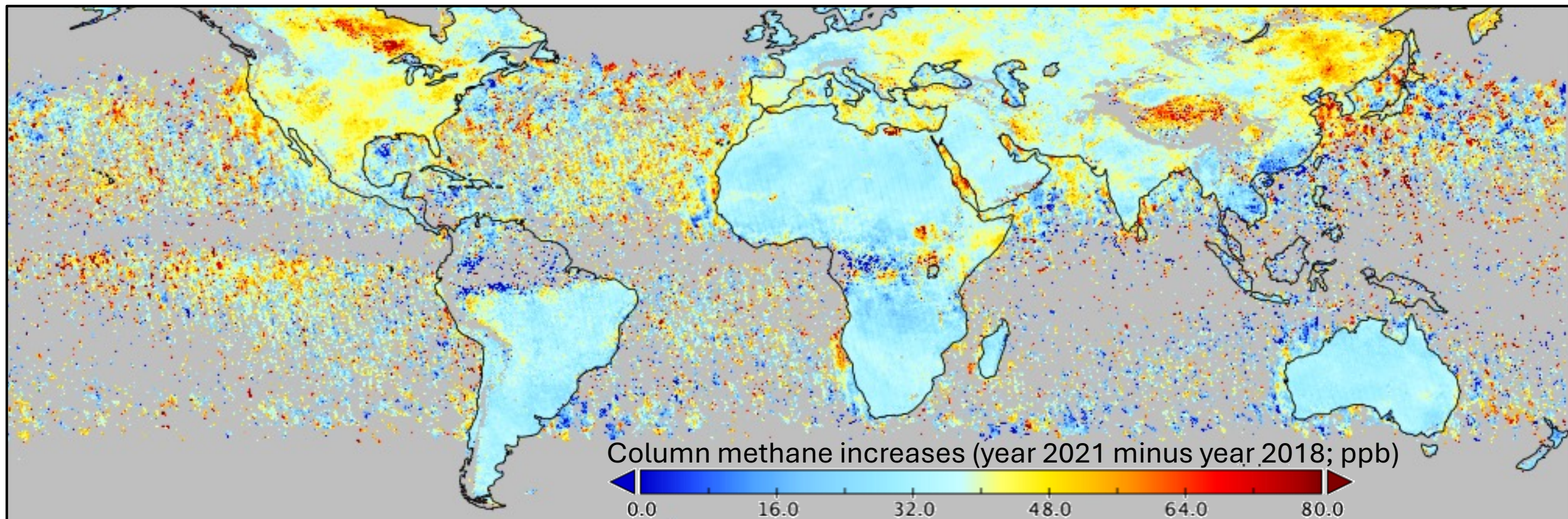
a) Time series of global-averaged methane column concentrations from TROPOMI satellite measurements (green: year 2019, black: year 2020, orange: year 2021). b) global emission underestimates, compared to bottom-up inventories, averaging from 2019 to 2021 (solid line: methane-isotopic joint inversion, dotted line with black points: methane-only inversion). c) 2019–2021 global methane emissions (solid line and segments with colors: methane-isotopic joint optimized emissions; dashed line: prior emission estimates from bottom-up inventories). Labeled in the bottom are underestimated sectoral sources and their dominant regions.

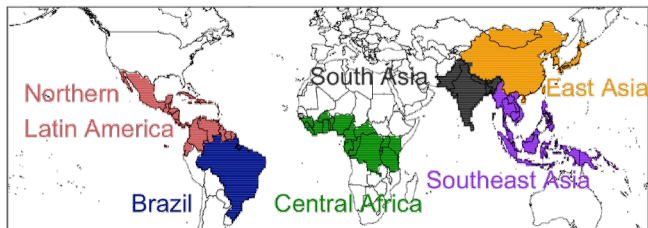
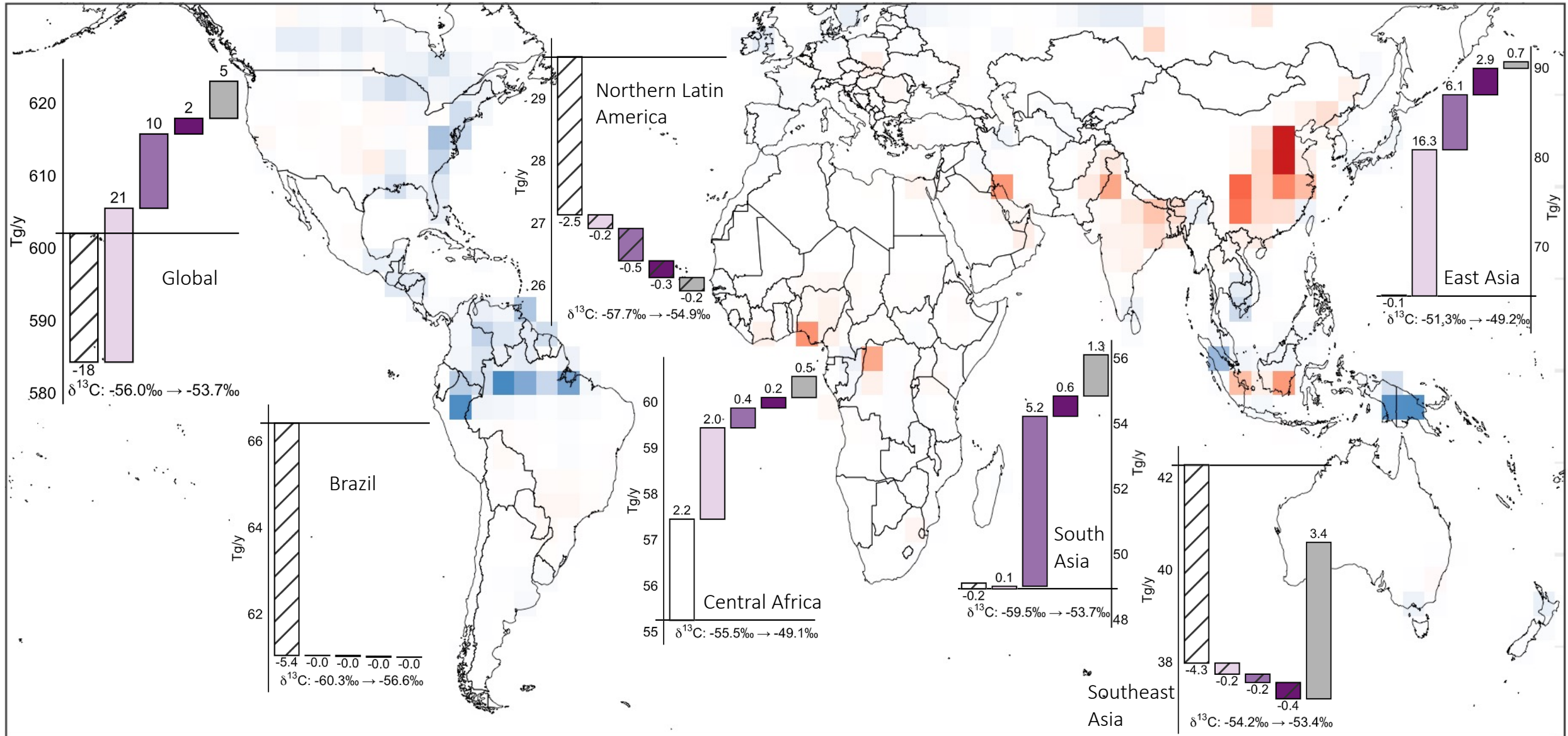
Editor's Summary:

Combining methane isotope measurements with satellite data shows recent methane growth is driven by stronger human emissions, with emissions redistributed toward tropical and subtropical regions, especially China and India.

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ARTICLE IN PRESS





Wetland

Other



Fossil

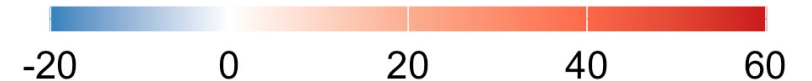
Agriculture

Waste



Increase

Decrease

kg day⁻¹ km⁻²

